

Quiz 1: Classification

2107541 (2026/1)
For individual student

Out: Due:

Sep Sep

9 16

2026 2026

23:59

 Submissions CANNOT be made after the due date/time.(16-Sep-2026 23:59)

 Submission can be made ONLY ONCE.

Instruction

undefined

My Work



Complete the question set

1 point

1

What is the main purpose of a classification model?

Pick a choice:

- ☐ Predict a continuous numerical value
- ☐ Predict a categorical class
- ☐ Reduce the number of features
- ☐ Calculate the correlation between variables

1 point

2

Which of the following is an example of a classification problem?

Pick a choice:

- ☐ Predicting daily rainfall in mm
- ☐ Predicting water temperature in °C
- ☐ Predicting whether water is potable or non-potable
- ☐ Predicting the tomorrow PM2.5 concentration

1 point

3

In a Decision Tree, what is the purpose of splitting the data?

Pick a choice:

- ☐ To increase the number of observations
- ☐ To create groups that are more homogeneous in their target class
- ☐ To remove all outliers from the the training data
- ☐ To normalize the distance between features and target class

1 point

4

Which algorithm is based on conditional probability ?

Pick a choice:

- ☐ KNN
- ☐ SVM
- ☐ Naive Bayes
- ☐ Decision Tree

1 point

5

Why is Naive Bayes called "naive"?

Pick a choice:

- ☐ It assumes all classes have equal probability
- ☐ It can only handle numerical data, can not for categorical data
- ☐ It does not require training data during the model training
- ☐ It assumes features are conditionally independent given the class

1 point

6

What function is commonly used in Logistic Regression to convert a linear output into a probability between 0 and 1?

Pick a choice:

- ☐ ReLU
- ☐ Sigmoid
- ☐ Softmax
- ☐ Hinge

1 point

7

Suppose a Logistic Regression model predicts:

$P(\text{Eutrophic})=0.82$

Using a threshold of 0.5, what is the predicted class?

Pick a choice:

- ☐ Non-eutrophic
- ☐ Eutrophic
- ☐ Cannot be determined
- ☐ Both classes

1 point

8

What is the main objective of a Support Vector Machine (SVM)?

Pick a choice:

- ☐ Find a decision boundary with a large margin between classes
- ☐ Find the nearest data point based on the calculated distance
- ☐ Calculate class probabilities using Bayes' theorem
- ☐ Build a tree structure by maximizing the information gain

1 point

9

In SVM, what are **support vectors**?

Pick a choice:

- ☐ The average point (centroid) of each class
- ☐ Data points farthest from the decision boundary
- ☐ Data points that have no effect on the model
- ☐ Data points that are critical in determining the decision boundary

1 point

10

In K-Nearest Neighbors (KNN), what happens when $K=3$?

Pick a choice:

- ☐ The model uses 3 random observations
- ☐ The model creates 3 classes
- ☐ The model considers the 3 nearest observations
- ☐ The model creates 3 decision boundaries with hard margin

1 point

11

Why is feature scaling particularly important for KNN?

Pick a choice:

- ☐ KNN uses distances between observations
- ☐ KNN uses Bayes' theorem
- ☐ KNN uses entropy and information gain
- ☐ KNN requires normally distributed features

1 point

12

A model produces the following confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	10	90

What is the **Recall**?

Pick a choice:

- ☐ 80%
- ☐ 85%
- ☐ 88.9%
- ☐ 90%

1 point

13

For harmful algal bloom (HAB) detection, why might **Recall** be an important metric?

Pick a choice:

- ☐ It measures how many predicted HAB cases are correct
- ☐ It measures how many actual HAB cases are successfully detected
- ☐ It measures the total number of observations
- ☐ It measures the training time of the model

1 point

14

A classification model has:

- TP = 90
- TN = 80
- FP = 20
- FN = 10

What is the **Accuracy**?

Pick a choice:

- ☐ 80%
- ☐ 85%
- ☐ 90%
- ☐ 95%

1 point

15

Which metric provides a balance between Precision and Recall?

Pick a choice:

- ☐ Accuracy
- ☐ ROC-AUC
- ☐ Specificity
- ☐ F1-score

(Optional) Call for the attention of

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